```
!pip install transformers datasets torch scikit-learn pandas matplotlib seaborn
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
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Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
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Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
```

```
import torch
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from datasets import load_dataset
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification,
    Trainer,
    TrainingArguments
)

from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix
```

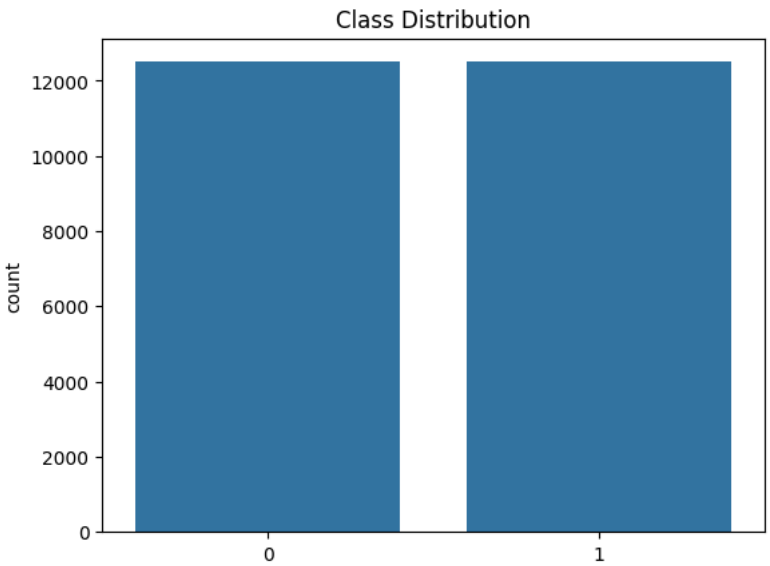
```
dataset = load_dataset("imdb")
dataset
```

```
README.md:          7.81k/?  [00:00<00:00,  374kB/s]

plain_text/train-00000-of-00001.parquet:  100%                    21.0M/21.0M  [00:00<00:00,  21.6MB/s]

plain_text/test-00000-of-00001.parquet:  100%                     20.5M/20.5M  [00:00<00:00,  47.7MB/s]

plain_text/unsupervised-00000-of-00001.p(…):  100%                42.0M/42.0M  [00:00<00:00,  50.5MB/s]

Generating train split:  100%                    25000/25000  [00:00<00:00,  79541.92  examples/s]

Generating test split:  100%                    25000/25000  [00:00<00:00,  76158.07  examples/s]

Generating unsupervised split:  100%                    50000/50000  [00:00<00:00,  137224.73  examples/s]
DatasetDict({
    train: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    test: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    unsupervised: Dataset({
        features: ['text', 'label'],
        num_rows: 50000
    })
})
```

```
print(dataset["train"][0])
```

```
{'text': 'I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that surrounded it when it was fir
```

```
labels = dataset["train"]["label"]
sns.countplot(x=labels)
plt.title("Class Distribution")
plt.show()
```




```
tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
config.json:  100%                    570/570  [00:00<00:00,  21.0kB/s]

tokenizer_config.json:  100%                    48.0/48.0  [00:00<00:00,  1.83kB/s]

vocab.txt:  100%                    232k/232k  [00:00<00:00,  1.90MB/s]

tokenizer.json:  100%                    466k/466k  [00:00<00:00,  14.3MB/s]
```

```
def tokenize_function(example):
    return tokenizer(
        example["text"],
        padding="max_length",
        truncation=True,
        max_length=128
    )

tokenized_datasets = dataset.map(tokenize_function, batched=True)
```

```
Map: 100%                                    25000/25000 [00:29<00:00, 744.08 examples/s]
Map: 100%                                    25000/25000 [00:27<00:00, 755.74 examples/s]
Map: 100%                                    50000/50000 [00:55<00:00, 855.49 examples/s]
```

```
tokenized_datasets = tokenized_datasets.remove_columns(["text"])
tokenized_datasets = tokenized_datasets.rename_column("label", "labels")
tokenized_datasets.set_format("torch")
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=2
)
```

```
model.safetensors: 100%                                    440M/440M [00:09<00:00, 105MB/s]
Loading weights: 100%                                    199/199 [00:00<00:00, 295.30it/s, Materializing param=bert.pooler.dense.weight]
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                         | Status     |
--------------------------------------------+------------+-
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.weight                | UNEXPECTED |
cls.predictions.bias                       | UNEXPECTED |
cls.predictions.transform.dense.bias       | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |
cls.predictions.transform.dense.weight     | UNEXPECTED |
cls.seq_relationship.bias                  | UNEXPECTED |
classifier.bias                            | MISSING    |
classifier.weight                          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    precision, recall, f1, _ = precision_recall_fscore_support(labels, predictions, average='binary')
    acc = accuracy_score(labels, predictions)
    return {
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    weight_decay=0.01,
    logging_dir="./logs",
)
```

```
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_datasets["train"].shuffle(seed=42).select(range(5000)),  # small subset for faster training
    eval_dataset=tokenized_datasets["test"].select(range(2000)),
    compute_metrics=compute_metrics
)
```

```
trainer.train()
```

[320/1250 59:42 < 2:54:37, 0.09 it/s, Epoch 0.51/2]

Epoch	Training Loss	Validation Loss

```
#results = trainer.evaluate()
#print(results)
```

```
#predictions = trainer.predict(tokenized_datasets["test"].select(range(2000)))
#preds = np.argmax(predictions.predictions, axis=-1)
#labels = predictions.label_ids

#cm = confusion_matrix(labels, preds)

#sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
#plt.xlabel("Predicted")
#plt.ylabel("Actual")
#plt.title("Confusion Matrix")
#plt.show()
```